

Mock Final Exam

CPSC 406 — Algorithm Analysis
 Chapman University · Spring 2026
 Instructor: Jonathan Weinberger

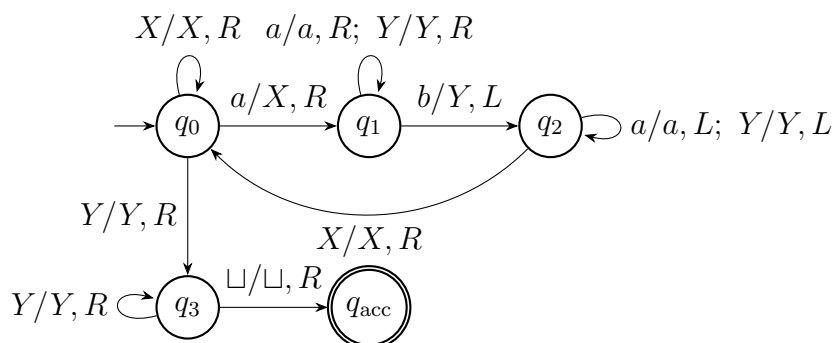
Time: 150 minutes

Resources: Four hand-written pages

Problem	1	2	3	4	5	6	7	8	Total
Max Points	5	5	5	5	5	5	5	5	40

1. (5 points) Turing Machines

Consider the following deterministic Turing machine over input alphabet $\Sigma = \{a, b\}$. The blank symbol is $\sqcup$. The start state is q_0 . The state q_{acc} is accepting, and omitted transitions go to a rejecting halt state.



For each word, determine whether the machine accepts or rejects it. Each correct row gives 1 point. A blank or wrong answer gives 0 points.

Word	Accepts	Rejects
<i>aabb</i>	☐	☐
<i>abb</i>	☐	☐
<i>abab</i>	☐	☐
<i>ab</i>	☐	☐
<i>aaabbb</i>	☐	☐

2. (5 points) **Landau Notation**

Let

$$g(n) = 4n^5 + 9n^2 \log n + 7\sqrt{n}.$$

For each of the following classes, determine whether $g(n)$ belongs to the class. Each correct row gives 1 point. A blank or wrong answer gives 0 points.

Class	Yes	No
$\mathcal{O}(n^4)$	☐	☐
$\mathcal{O}(n^5)$	☐	☐
$\mathcal{O}(n^6)$	☐	☐
$\mathcal{O}(2^{\sqrt{n}})$	☐	☐
$\mathcal{O}(n^5 \log n)$	☐	☐

3. CNF and Satisfiability

- (a) (2 points) Convert the following formula into conjunctive normal form (CNF). Show the main steps of your conversion.

$$\varphi = (p \rightarrow (q \vee r)) \wedge \neg(q \wedge r).$$

- (b) (3 points) Determine whether φ is satisfiable. If it is satisfiable, give a satisfying assignment.

4. Logical Modeling

A printer system uses the following policy for allowing a print job.

1. If the printer is in maintenance mode, then printing is not allowed.
2. If the printer is online and has paper, then printing is allowed.
3. If an administrator override is active, then printing is allowed.

A system log reports:

- the printer was in maintenance mode,
 - the printer was online,
 - the printer had paper,
 - no administrator override was active,
 - printing was allowed.
- (a) (2 points) Formalize the policy and incident log using propositional logic. Clearly define your propositional variables.
- (b) (1 point) Convert your formalization into a set of clauses.
- (c) (1 point) Use resolution to determine whether the log is consistent with the policy.
- (d) (1 point) If it is inconsistent, identify the conflict.

5. (5 points) **Induction and Big-O Bounds**

Consider a function $T : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ satisfying

$$T(1) = 1 \quad \text{and} \quad T(n) \leq 2T\left(\frac{n}{2}\right) + n \quad \text{for all powers of two } n \geq 2.$$

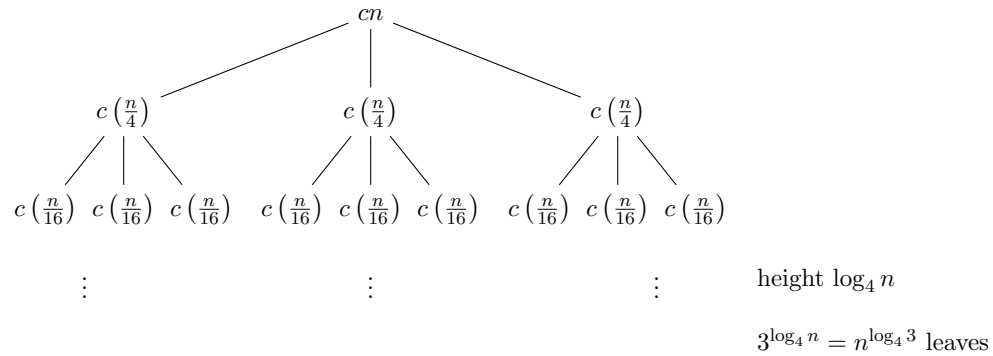
Use induction to prove that

$$T(n) \leq n \log_2 n + n \quad \text{for all powers of two } n \geq 1.$$

Conclude that $T(n) \in \mathcal{O}(n \log n)$.

6. Recurrence Trees

The following recurrence tree represents the work done by a recursive algorithm on an input of size n .



- (a) (2 points) Devise a recurrence equation for the worst-case runtime represented by this recurrence tree.
- (b) (3 points) Determine a Big-O bound for this recurrence. You may use the Master theorem or read the bound directly from the recurrence tree.

7. (5 points) **Random Permutations and Indicator Variables**

Let $A[1..n]$ be a uniform random permutation of $1, 2, \dots, n$. Let k be an integer with $1 \leq k \leq n$.

Use indicator random variables to compute the expected number of positions i such that $A[i] \leq k$.

8. Probabilistic Algorithm Analysis

Consider the following randomized algorithm.

```
COUNTSIXES( $n$ )  
   $c \leftarrow 0$   
  for  $i \leftarrow 1$  to  $n$  do  
    for  $j \leftarrow 1$  to  $i$  do  
       $r \leftarrow$  a uniformly random integer from  $\{1, \dots, 6\}$   
      if  $r = 6$  then  $c \leftarrow c + 1$   
  return  $c$ 
```

- (a) (2 points) Give a Big-O bound for the worst-case running time. Briefly justify your answer.
- (b) (3 points) Give the expected value of the returned number c . Briefly justify your answer.